DATA

Pie Charts, Line and Bar Diagrams

eBook

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DI – Line, Bar & Pie E-Book

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1.1 Line Graphs

A line graph or line chart, is a type of chart used to visualize the value of something over a period of time. The line graph therefore helps to define the relationship between two sets of values; with one data set is always being dependent on the other set. It is commonly used to display change over time as a series of various data points connected by straight line segments on two axis. In competitive exams you will see a set of questions based on line graph problems, in which you are supposed to do graph interpretation, analyze the information and then answer the questions.

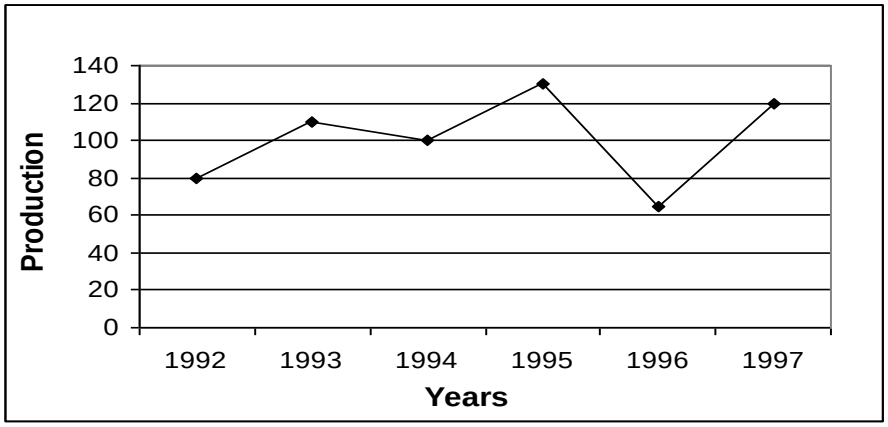
How to make a line graph: The line graph consists of a horizontal x-axis and a vertical y-axis. Line graphs are drawn so that the dependent data are on the vertical y-axis and the independent data are on the horizontal x-axis (e.g. time). Most of the line graphs only deal with positive numbers values, so these axis typically intersect near the bottom of the y-axis & the left end of the x-axis. The point at which the axis intersect is usually (0, 0). Each axis is labeled with a specific data type. E.g., the x-axis could be days, quarters, weeks, or years, while the y-axis could be showing revenue, production, expenditure, cost in rupees etc.

Line graphs are useful as they show data variables & trends very clearly. Line graphs can help to make predictions about results of data not yet recorded. This type of graph is most powerful visual tool for finance, marketing, & other areas. It is also useful in weather monitoring, laboratory research, or any other function involving a correlation b/w two numerical values. If more than one line is on the chart, it can be used as a comparison between them. When comparing these data sets, line graphs are only applicable if the axis follows the same scales. In the examples given below you will learn about interpretation of graphs based on lines.

Note: *Slope is the most important observation in this case. It helps in comparing the magnitude of change between any two consecutive points on graph. E.g. Steeper the slope, greater is the change in magnitude between two consecutive points.*

1.2 Line Graph Examples

DIRECTIONS: Given below are line graph Examples with questions, which show the annual food grain production from 1992 to 1997. Refer to the graph & answer the questions based on line graph as given below.



Ex. 1: How much is the approximate percentage increase in production from 1992 to 1993?

1. 87.5% 2. 37.5% 3. 12.5% 4. None of these

Sol: Here we look up the values of the production for the 2 years first. Locate 1992 on the X axis, which denotes years. Move vertically up along the Y axis direction in 1992, and we get value of production in 1992 as 80. In the same way we get the value of production in 1993 as 110.
In calculating % increases & decreases. In this case 1992, it is very important to remember that the original year is the one that is used as the reference year,. First we calculate the absolute increase which is 110 – 80 = 30. Now we have to express 30 as a percentage of the production in 1992, which is 80. So the required answer is 100 × 30/80 = 37.5%. Hence answer is (2).

Ex. 2: In which year was the production equal to the average of that of 1997 and 1992?

1. 1996 2. 1993 3.1995 4. 1994

Sol: The production in 1992 was 80, that in 1997 was 120.
The average production of 1992 and 1997 (80 + 120)/2 = 100.
This was the production in 1994. Hence Answer is (4).

Ex. 3: In how many cases was the production below the average of the given years?

1. 1 2. 2 3.3 4. 4

Sol: Average production for 6 years = $\frac{80+110+130+100+120+65}{6}$ = 101

Clearly in 1992, 1994 and 1996, the production was less than average. Hence answer (3)
A small digression over here. We can also calculate the average by using a small statistical property which states that the median and average are typically very close for most normally distributed data. To calculate the median value in graphical data, use the edge of a paper and move it upwards till only half the data points are visible. The intersection of the edge of the paper with the Y axis is the median value – and approximately the average.

Ex. 4: If production in 1998 is estimated to be 25% lower than that in 1997, the estimate of 1998 production is?

1. 120 units 2. 100 units 3.104 units 4. 90 units

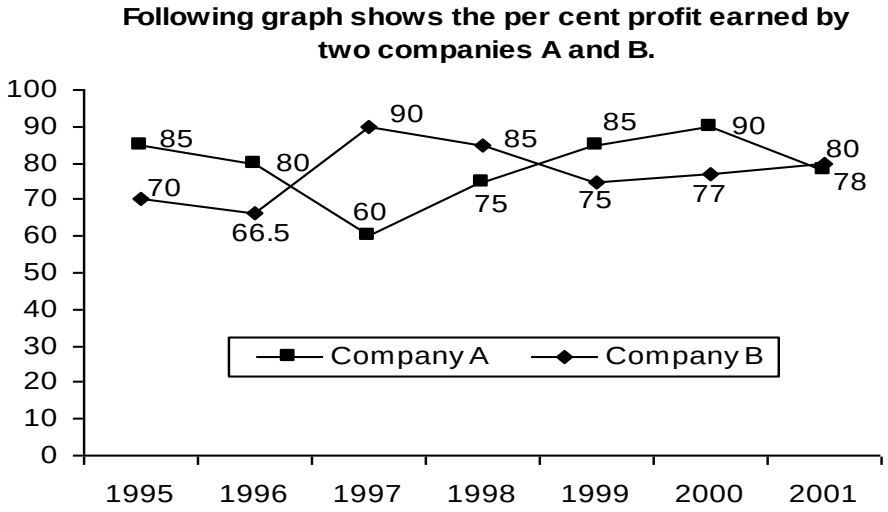
Sol: The production in 1997 is 120 units. 25% of 120 = 30 units less. 120 – 30 = 90 units is the new production. Hence answer is (4).

Ex. 5: The maximum increase in food grain production has been in

1. 1997 2. 1996 3.1995 4. 1994

Sol: This is a question where graphical data can really be interpreted well without any calculations. We call this visual inspection. So what do we inspect?
Here remember that absolute increase has been asked. The difference in Y coordinates of the adjacent years will give us the increase. Since the distance between consecutive years is the same on the X axis, we can check the slopes of various segments. By visual inspection, maximum increase in food grain production has been in the year 1997 (55 units by calculation).

DIRECTIONS ***for questions 6 - 10:*** *Study the following graph carefully and answer accordingly: Following graph shows the per cent profit earned by two companies A and B.*



Ex. 6: Income of company B in 2000 was Rs.1239 lakhs. What was the investment in that year (in Rs lakhs) of company B?

A. 700 B. 800 C. 650 D. 193.03

Sol: Investment of company A in 2000 = 1239 x $\frac{100}{177}$ = Rs.700 lakh

Ex. 7: Investment of company B in 1998 was 20% more than that in the previous year. Income in 1998 of company B was what per cent of its income in 1997?

A. 10% B. $102\frac{2}{3}\%$ C. $106\frac{2}{3}\%$ D. None of these

Sol: Let us assume the investment of Company B in 1997 is Rs. 100. Investment of company B in 1998 was 20% more than that in the previous year.
So that, the investment of Company B in 1998 is Rs. 120
Income in 1997 of company B= 90 % of 100 = Rs. 90
Income in 1998 of company B= 85% of 120 = Rs. 102. Required % = $\frac{102}{90}\times 100 = 113\frac{1}{3}\%$

Ex. 8: In which of the following years the ratio of investment and income is maximum for company A?

A. 2001 B. 1995 C. 1998 D. 2000

Sol: Quicker Method: The ratio of investment and income will be maximum when percentage profit is minimum. Therefore, The ratio of investment and income will be maximum in 2001.

Ex. 9: If the income of company B in 1996 was same as the income of company B in 1998, what would be the ratio of investment of company B in 1998 to the investment of company B in 1996?

A. 9 : 10 B. 10 : 9 C. 13 : 15 D. 15 : 13

Sol: Required ratio $= \frac{100}{185} \times \frac{166.5}{100} = \frac{333}{370} = \frac{37 \times 9}{37 \times 10} = 9:10$

Ex. 10: In which of the following years the ratio of investment and income is maximum for company B?

A. 1995 B. 1999 C. 2000 D. 1998

Sol: Quicker Method: The ratio of investment and income will be maximum when percentage profit is minimum. Therefore, The ratio of investment and income will be maximum in 1995.

2.1 Bar Graph Concepts

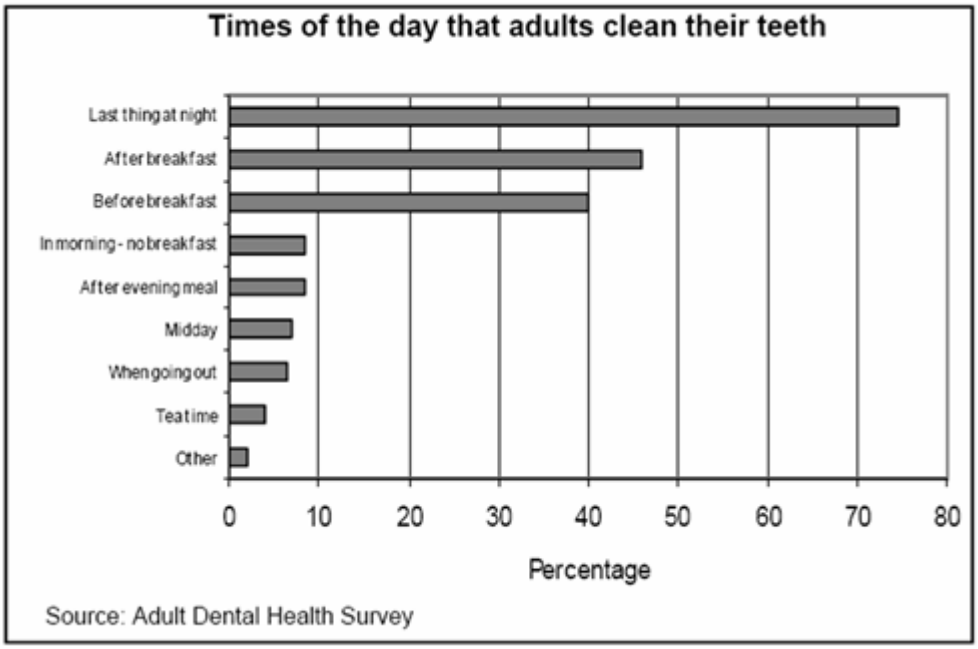
A **Bar Graph** or Chart is a chart that uses horizontal or vertical bars to show comparisons between categories of data. It will have two axes. One of the axes will describe the categories being compared, and the other axes will represent the values of the data. It does not make a difference, which axis is representing what, but it will make it clear what the bar graph is showing. If the descriptions are given horizontally, the bars will be oriented vertically, and if the values are at the represented vertically the bars will be oriented horizontally.

Bar chart is one of the easiest, attractive and commonly used methods of presenting data in graphs. Questions based on these charts are commonly asked in competitive exams. Further, in these Charts, the questions on percent change and weighted average based questions are generally asked in exams.

Different types of bar charts & How to make a Bar Graph

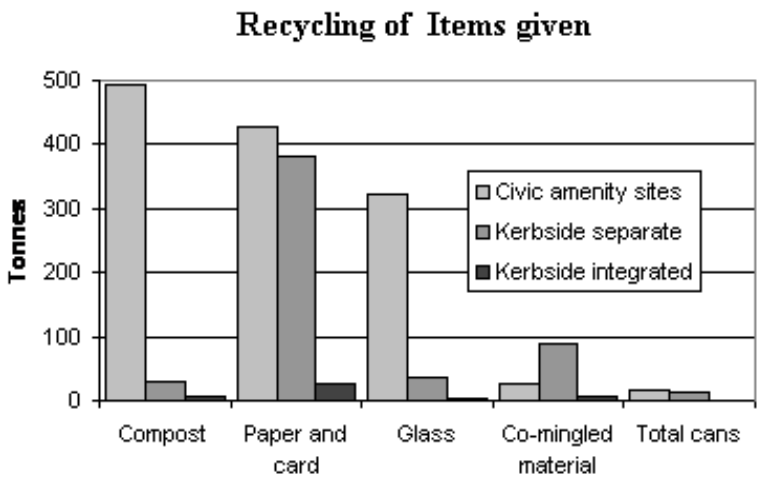
Horizontal charts

Bar charts are usually drawn with vertical bars which implies that the taller the bar, the larger the value. However, it is also possible to make horizontal bar charts which means that the longer the bar, the larger the data. This becomes particularly effective when the different categories have long titles that would be difficult to mention below a usual vertical bar, or when there are a higher number of different items and there is inadequate area to accommodate all the columns required for a vertical chart in the page.



Grouped charts

This graph shows information about different sub-groups of the main items. In the graph given below, a grouped bar chart has been used to show the different sub-groups of household materials which are reused. These charts are used to show several sub-groups of each item but it needs to be taken care that the chart does not contain too much information, which becomes difficult to interpret. These charts can be drawn vertically as well as horizontally depending upon the nature and values of the data to be presented.

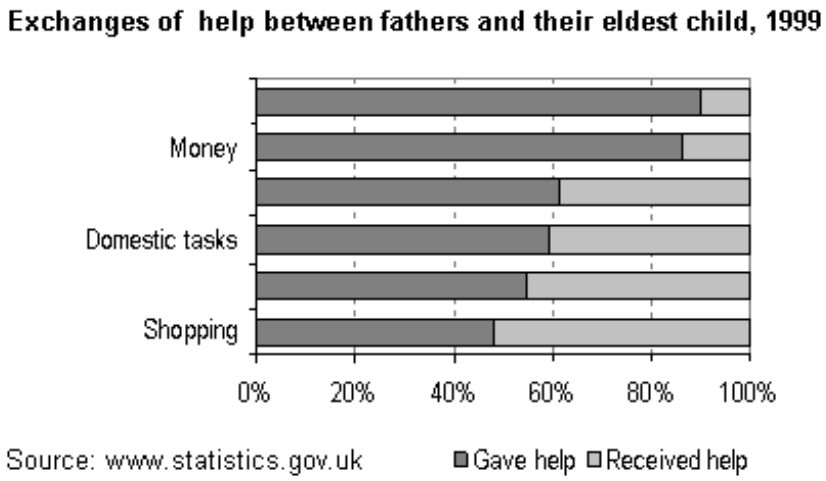


Source: Department of Environment, Transport and the Regions

Stacked bar charts

In these bar charts, the bars representing the other groups are placed on top of each other to make a single graph. The overall length or height of the bar tells the total size of the category while different shades or size are used to display the relative proportion of the different sub-categories. You will see these kinds of graphs in so many exams, so those are really important.

These stacks charts can be used to show the proportionate contribution of different sub-categories, which contribute to each separate category as well. In such case the bars representing the different categories are all of the same size.

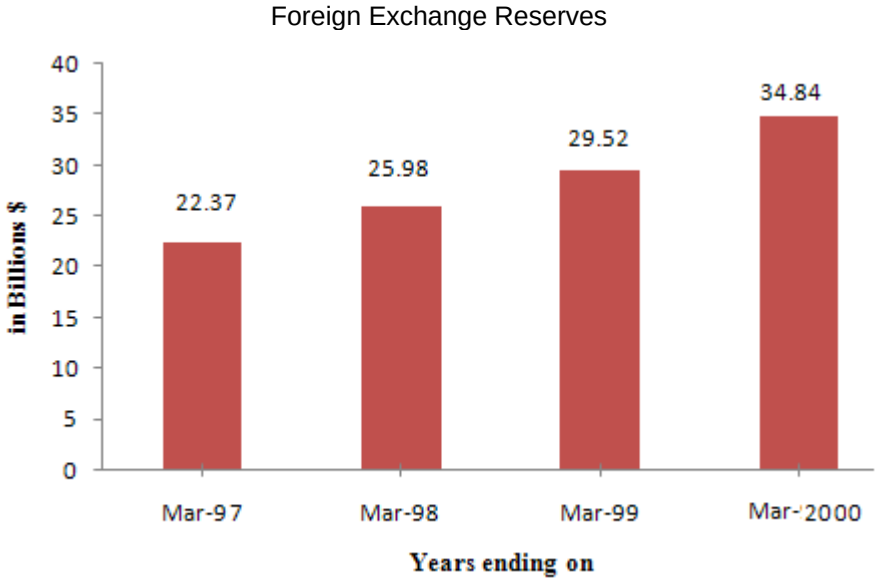


Source: www.statistics.gov.uk

Before attempting any questions on bar graphs, you must go through the following bar graph Examples and questions. You must attempt 40 – 50 questions of data interpretation before solving the question on the same in the actual examination.

2.2 Bar Graph Examples

DIRECTIONS *for questions:* *Analyse the following graph and answer the questions given below.*



Ex.1: The foreign exchange reserve in March 1998 increased by what percent as compared to the corresponding month last year?

A. 13.9% B. 21.7% C. 11.8% D. 16.1%

Sol: Foreign exchange reserve in March '98 = 25.98 bn.

For corresponding period last year i.e. March '97, Foreign exchange reserve = 22.37 bn.
Increase in foreign exchange reserve = 3.61 bn.
Percentage increase = 100 × 3.61/22.37 = 16.13%.

Besides this, it can also be seen that 10% of the base i.e. 22.37 is 2.2 and 15% will be 3.3. Hence, the answer has to be slightly greater that 15%. Hence answer will be fourth option.

Ex.2: The foreign exchange reserve in March 1997 was - - - - % less than what was in March 1998.

A. 16.2% B. 10.2% C. 13.9% D. 19.8%

Sol: Foreign exchange reserve in March '98 = 25.98 bn. Foreign exchange reserve in March '97 = 22.37bn. Decrease in foreign exchange reserve = 3.61bn. Percentage decrease = 100×3.61/25.98 = 13.9%.
Besides that, it can also be seen that 10% of the base i.e. 25.98 is approximately 2.6 and 15% will be 3.9. Hence, the answer has to be more than 10% and less than 15%. Hence, the third option is the answer.

Ex.3: The average rate of increase per year of foreign exchange reserve in India from March 1997 to March 1999 was approximately

A. 10.5% B. 16% C. 26% D. 32%

Sol: Foreign exchange reserve in March'97 = 22.37 bn.
Foreign exchange reserve in March '99 = 29.52 bn.
Increase in foreign exchange reserve = 7.15 bn.
Percentage increase in two years = 100× 7.15/22.37 = 32%.
Average rate of increase per year = 32/2 = 16%.

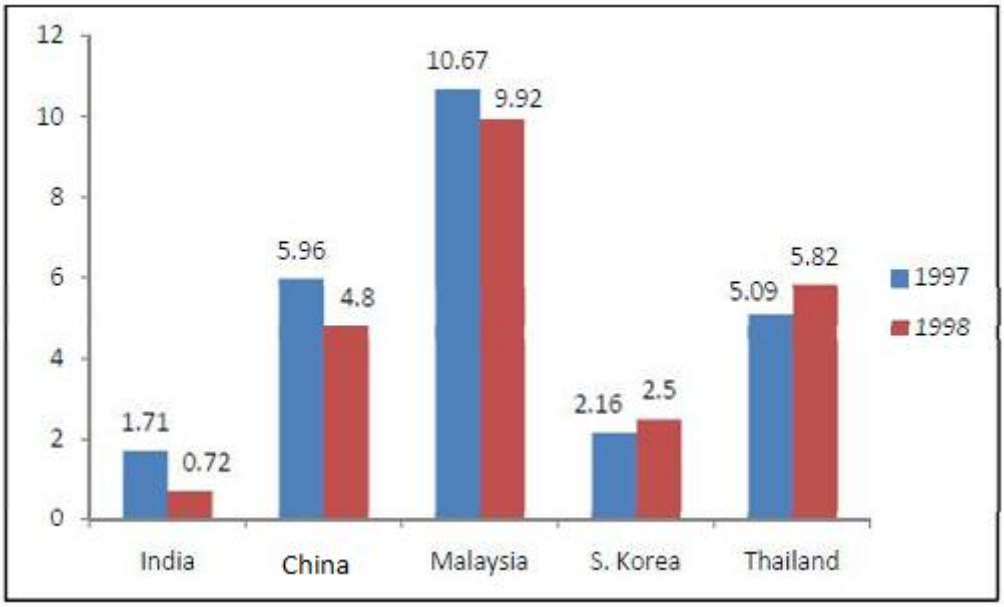
It can noted here that the increase of 7.15 is roughly one-third of 22 i.e. 33.33%. For every year it will become 33/2 = 16.5%. Hence, second option is the answer.

Ex.4: What was the average foreign exchange reserve for the three years i.e. March 1997, March 1998 and March 1999, approximately?

A. $ 26 bn B. $ 22.18 bn C. $28.88 bn D. $ 12.56 bn

Sol: From questions 1 and 2, the difference between March '97, '98 and '99 is almost same. Thus, the average would be the middle value i.e. $ 25.98 Billion = $ 26 billion.

DIRECTIONS: The following graph gives the data about Foreign Equity Inflow (FEI) for the five countries for two years 97 and 98.FEI is taken as the ratio of foreign equity inflow to the country's GDP, which is expressed as a percentage in the following graph.



Ex. 5: The country with the highest percentage change in FEI in 1998 relative to its FEI in 1997, is

A. India B. China C. Malaysia D. Thailand

Sol: The common mistake that students make while solving this problem is that they just take the difference in the figures given on the bar tops as these are given in percentages. As per that the answer is China which has the maximum difference of 5.96 – 4.8 = 1.16 but this is technically wrong as the question is to answer the percent change from the year 1997 to 1998 for the countries and not the change only. You can solve the question by calculating the percent change for all the given countries and mark the answer. Besides that a short cut can be used as well, just visualize the bars of the two years given. You can see that the graph for 1998 is not even half the height of the graph in 1997 for India, that means the decrease is more than 50% and no other country is closer to that even. Hence, the answer is India.

Ex. 6: Which country has maximum change in the FEI?

A. India B. China C. Malaysia D. Thailand

Sol: In this question, you are only asked to identify the country with the maximum change and not the percent change. You can calculate this by using the difference between the values given on the top. So, as per this, the answer is China as it has the maximum difference of 1.16.

Here you need to apply the concept of subtraction to find the answer.

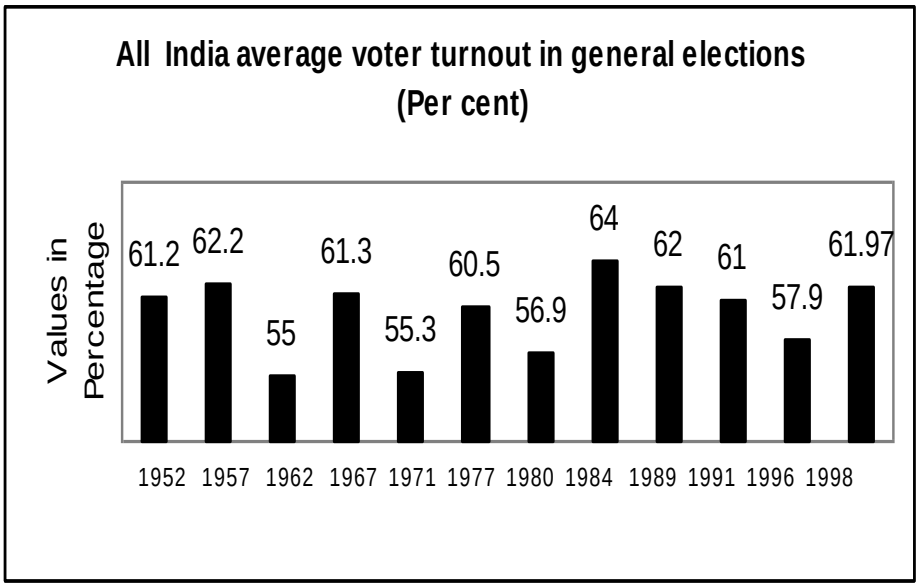
Ex. 7: Supposing in Thailand 20% of FEI in 1997 and 50% of FEI in 1998 goes to Education sector, then find the ratio of the amounts allocated to Education in 1997 to 1998, assuming the GDPs of both of these years for Thailand to be same.

A. 1 : 3 B. 5 : 21 C. 6 : 11 D. 7 : 25

Sol: Taking the data from the chart for Thailand you can see that in the year 1997, the FEI is 5.09% of GDP and 20% of it is allocated to the Education sector and in year 1998, 5.82% of the GDP is FEI and 50% of it goes to the Education sector. Now if you try to find it by the conventional method, it will take a lot of time and energy. Now as you have to find the ratio of amount on Education sector in the 1997 and 1998, you can do rough calculations by taking 5.09% as 5% for 1997 and 5.82% i.e. roughly 6% approx. for 1998. After that just calculate ratio as 5 × 20 × GDP : 6 × 50 × GDP = 100 : 300 = 1 : 3 approximately.

DIRECTIONS *for questions 8 - 12:* *Study the following graph and answer the questions given below it:*

EXERCISING FRANCHISE



Ex. 8: The difference between highest and lowest ever voters' turn- out is

A. 8% B. 3% C. 7% D. 9%

Sol: The difference between highest (64%) and lowest (55%) voters turn out is = 64% - 55% = 9%

Ex. 9: In which of the following consecutive elections, the fluctuation in voters' turnout (in percentage) was the highest?

A. 1957-62 B. 1984-89 C. 1996-98 D. 1971-77

Sol: Option (1) The required fluctuating = 62.2 – 55 = 7.2%
Option (2) The required fluctuating = 64 – 62 = 2%
Option (3) The required fluctuating = 61.97 – 57.9 = 4.07%
Option (4) The required fluctuating = 60.5 – 55.3 = 5.2%, Obviously Option (1) satisfies

Ex. 10: Between1952 to 1998, the average voter turnout was

A. 56.8% B. 40.9% C. 62.7% D. 60.5%

Sol: The average voter turnout between 1952 to 1998

$$= \frac{61.2+62.2+55+61.3+55.3+60.5+56.9+64+62+61+57.9+61.97}{12} = 59.93\% \approx 60.5\%$$

Ex. 11: In which of the following elections, the growth in the average voter turnout was the highest in comparison to the previous election?

A. 1967 B. 1984 C. 1998 D. 1977

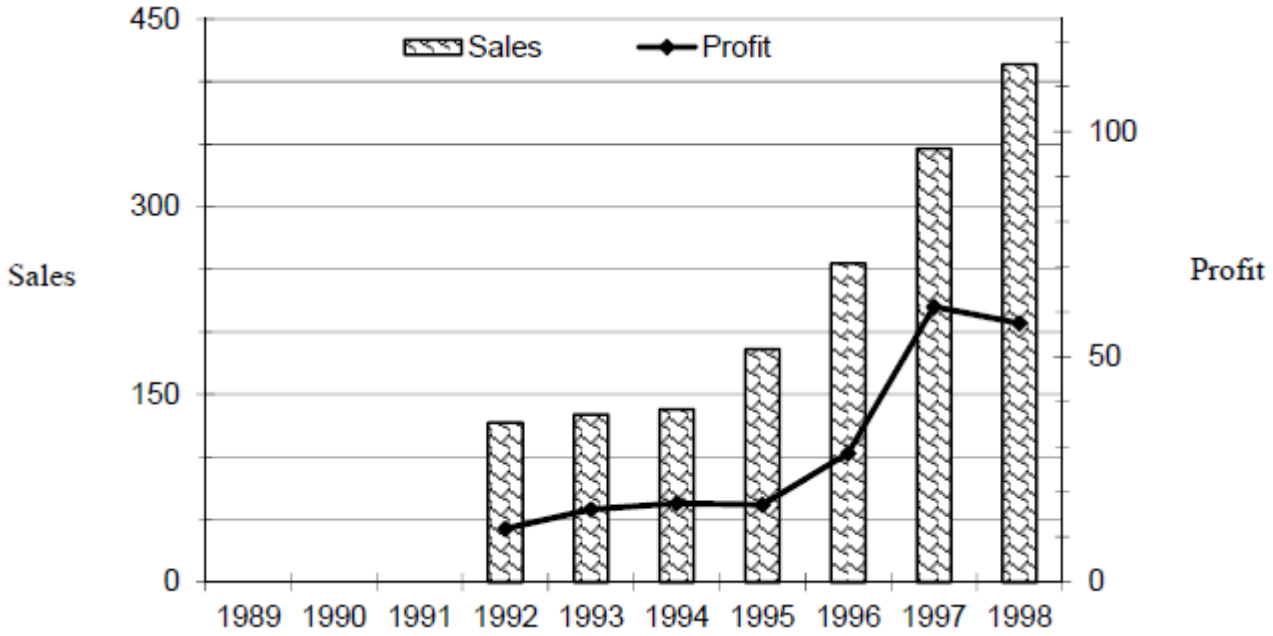
Sol: For 1984, it is highest (56.9 to 64) = 7.1%

Ex. 12: The lowest average voters' turnout in comparison to the total voters was noticed in

A. 1971 B. 1996 C. 1962 D. 1952

Sol: Visual observation reveals it is for year 1962

DIRECTIONS *for questions 13 – 15:* Refer to the following graph of sales and profit figures of ABC Ltd Answer the questions that follow.



Ex. 13: Compounded average growth rate for profit was highest in the period

A. 1994-96 B. 1995-97 C.1996-98 D.1995-98

Sol: By observing growth trend.Compounded average growth rate for profit was highest in the period of 1995-97.

Ex. 14: Return on sales (Profit/sales) was highest in

A. 1995 B. 1996 C.1997 D.1998

Sol: Return of sales in 1995 = $\frac{20}{200}$ or $\frac{1}{10}$;

Return of sales in 1996 = $\frac{30}{270}$ or $\frac{1}{9}$

Return of sales in 1997 = $\frac{60}{360}$ or $\frac{1}{6}$;

Return of sales in 1998 = $\frac{55}{410}$ or $\frac{1}{7.5}$

So, highest return comes in 1997.

Ex. 15: Return on sales (profit/sales) exceeded 15%

A. Once B. Twice C. Thrice D. Never

Sol: Return of sales in 1995 = $\frac{20}{200}$ or $\frac{1}{10}$;

Return of sales in 1996 = $\frac{30}{270}$ or $\frac{1}{9}$

Return of sales in 1997 = $\frac{60}{360}$ or $\frac{1}{6}$;

Return of sales in 1998 = $\frac{55}{410}$ or $\frac{1}{7.5}$

Return on sales exceeds 15% only in 1997.

Ex. 16: Profit growth exceeding 50% (over previous year) was registered

A. Once B. Twice C. Thrice D. Never

Sol: Profit growth exceeds 50 % in 1996 i.e. from 20 to 30 and in 1997 from 30 to 60.

Ex. 17: In how many years did the sales growth exceeded 25% of that of previous year :

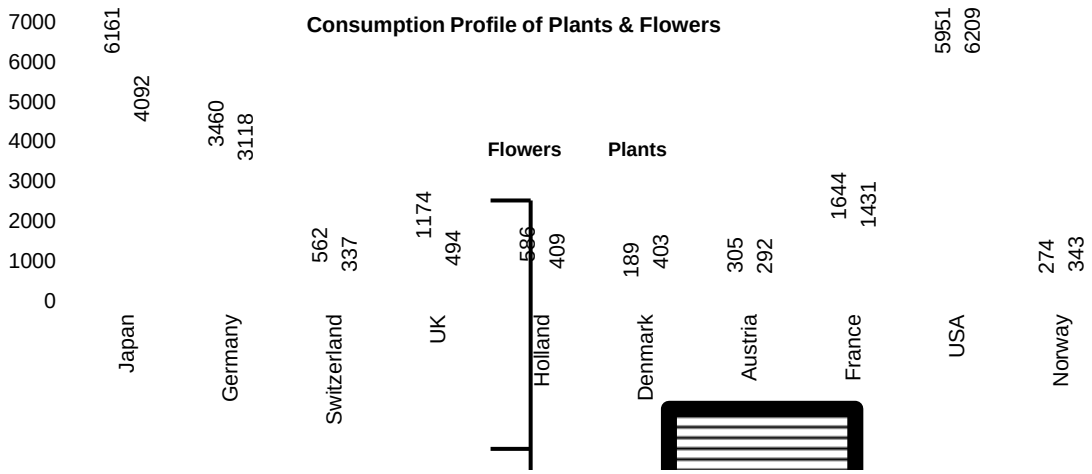
A. Never B. One year C. Two years D. Three years

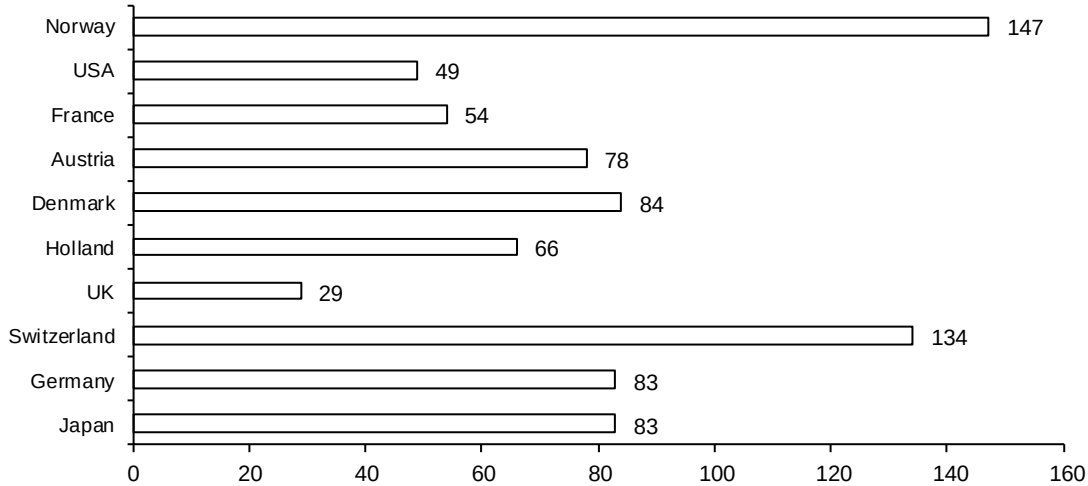
Sol: Sales growth exceeds 25% in 1995, 1996 and 1997.

DIRECTIONS *for questions 18 – 20:* *Refer to the bar charts given below and answer the questions that follow.*

First chart gives the consumption of flowers and plants in US Dollar millions.

The second Bar Graph gives the Per Capita Consumption of Flowers and Plants (in US dollars).





Ex. 18: What is the total consumption of plants in all the countries given ($ million)?

1. 20000 2. 18000 3.17000 4.15000 5. 27000

Sol: Total Consumption of plants (4092 + 3118 + 337 + 494 + 409 + 403 + 292 + 1431+ 6209 + 343) = 17128. Hence the answer is (3).

Ex. 19: What could be the approximate population of Japan?

1. 125 million 2.200 million 3.300 million 4.250 million 5.Cannot be determined.

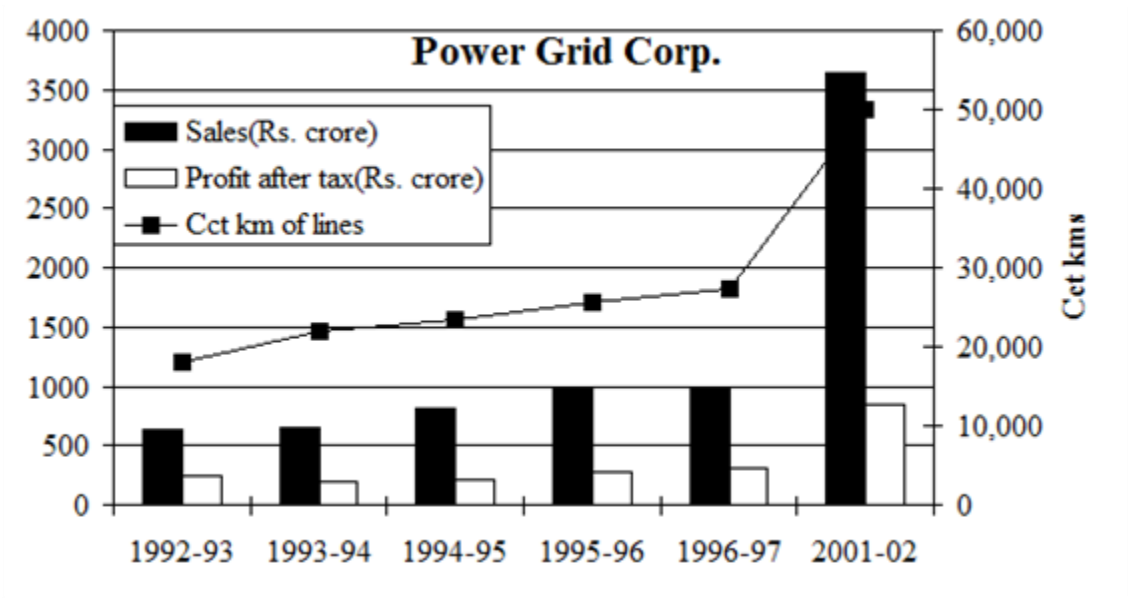
Sol: Total consumption of plants and flowers in Japan = 6161+4092 =10253 million.
Hence per capita consumption = 83.
Population = 10253/83 million =123 million.
Hence the answer is (1).

Ex. 20: By what percent is the plant consumption more or less than flower consumption?

A. 28 % more B. 28 % less C.18 % more D. 8 % less E. 18% less

Sol: Plant consumption = 4092 + 3118 + 337 + 494 + 409 + 403 + 292 + 1431 + 6209 + 343 = 17128. Flower consumption = 6161 + 3460 + 562 + 1174 + 586 + 189 + 305 + 1644 + 5951 + 274 = 20306. % of Plant consumption less than flower consumption = ((20306 – 17128)/17128) × 100 = 18.58%. Hence the answer is (5).

DIRECTIONS *for questions 21 to 24:* *Refer to the following table which is based on Power Grid*



Corporation.

Ex. 21: The revenue generated per circuit kilometer was the lowest in

A. 1992-93 B. 1993-94 C.1995-96 D.1996-97 E. 2001-02

Sol: Between 92-93 and 93-94 – revenue is almost same, but circuit km went up. So the lower of the two is 93-94 = 600 /23000. Similarly for 96-97 = 1000 / 27000.
Comparing 93-94 is the lowest. Hence 2

Ex. 22: The highest average annual growth rate in revenues is

A. 250% B. 2.5% C. 25 % D. 50% E. 75 %

Sol: Because the 5 year growth has been 250%, annual growth = 50% .Hence 4.

Ex. 23: Power Grid Corporation has drawn up a plan to invest Rs 13,200crore in the period 1996-97 to 2001-02. Assuming that all this investment is to be in increasing circuit km of transmission lines, the anticipated investment required for one km of transmission circuit is

A. Rs 1.8 cr. B. Rs 0.55 cr. C. Rs 0.46 cr. D. Rs. 0. 50 cr E. Rs. 0.65 cr.

Sol: Investment is to be in increasing circuit km of transmission lines is = 13200 / 24000 = 0.55.Hence 2.

Ex. 24: Profitability (Profit / Sales) is the highest in

A. 1992-93 B. 1993-94 C. 2001-02 D. 1996-97 E. 1995 – 96

Sol: PAT/Sales for 92-93 = 200/600.
For 2001-02 = 750/3500. Between 92-93 and 93-94 is higher for 92-93. It cannot be 96-97.
Hence 1.

2.3 Use of Weighted Average in Graph

In this article, you will learn the application of concepts of average and weighed average in data interpretation. You already know that the average or arithmetic mean is the sum of all the terms divided by the number of terms. Basically, in average, each of the data points contributes equally to the final average.

The **weighted arithmetic mean** is similar to arithmetic mean (the most common type of average), except that, in the former, each data point does not contribute equally to the final average. In weighted arithmetic mean, some data points contribute more than others, as they have more weightage or they appear more frequently.

The weightage average is calculated as:

$$x = \frac{x1(w1) + x2(w2) + x3(w3) + \cdots}{w1 + w2 + w3 + \cdots}$$

$$x = \frac{x1(w1) + x2(w2) + x3(w3) + \ldots}{w1 + w2 + w3 + \ldots}$$

Ex.1: Ram wants to buy a new cell phone, and he gives the following weightage to different features while buying: Camera Quality= 50%; Battery Life= 25%; Processor= 25%. He accessed the most trusted cell phone rating website on Internet and found:

i) The Sony J gets 6 (out of 10) for Camera Quality, 7 for Battery Life and 8 for Processor.
ii) The Samsung S gets 9 for Camera Quality, 7 for Battery Life and 4 for Processor.

Which cellphone should he buy?

Sol: Here, had he given equal weightage or priority to the 3 features, the simple arithmetic mean, would be equal to (6 + 7 + 8)/3 = 7 for Sony and (9 +7 +4)/3 = 6.67 for Samsung. In this case, he should buy Sony J.

But, in this case the importance attached to different features is unequal. So, you need to calculate the weighted average for both. The weighted average in this case would be found as

$$\frac{\text{weightage of camera} \times \text{rating} + \text{weightage of battery} \times \text{rating} + \text{weightage of processor} \times \text{rating}}{\text{TotalWeightage}}$$

Weighted average of Sony J = 0.5(6) + 0.25(7) + 0.25(8) /.5 + .25 + .25 = 6.75

$$\frac{0.5(6) + 0.25(7) + 0.25(8)}{.5 + .25 + .25} = 6.75$$

Weighted average of Samsung S= 0.5(9) + 0.25(7) + 0.25 (4)/.5 + .25 + .25 = 7.25

$$\frac{0.5(9) + 0.25(7) + 0.25(4)}{.5 + .25 + .25} = 7.25$$

So, he should buy Samsung S.

Also, in a weighted average based question, you can adopt two approaches i) Take the weights given as numbers, or ii) Convert them into their simplest form and take the weights accordingly.

In the above Ex., the preference or priority or weights are given as 50%, 25% & 25%. They can be reduced to the lowest possible terms i.e. 2,1 and 1 for camera, battery and processor respectively (dividing weights by the same number i.e. 25 in this case, to convert in simplest form)
Weighted average for Sony J = 2(6) + 1(7) + 1(8) / 2 + 1 + 1 = 6.75

$$\frac{2(6)+1(7)+1(8)}{2+1+1}=6.75$$

Weighted average for Samsung S= 2(9) + 1(7) + 1(4) / 2 + 1 + 1 = 7.75

$$\frac{2(9)+1(7)+1(4)}{2+1+1}=7.25$$

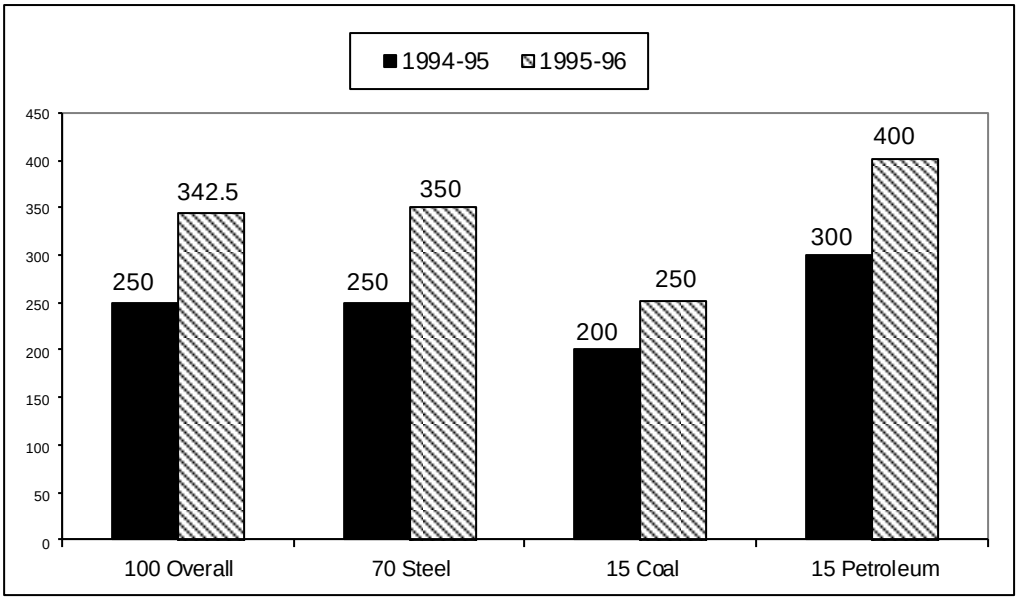
Ex. 2: Ram is a sales executive and his preference while buying a cellphone is a good battery life. His priority in percentages are, Camera Quality= 20%, Battery Life= 50% and Processor= 30%

Sol: The weights are 2, 5 and 3 (for camera, battery and processor, respectively)
Weighted average for Sony J = 2(6) + 5(7) + 3(8)/2 + 5 +3 = 7.1 $\frac{2(6)+5(7)+3(8)}{2+5+3}=7.1$

Weighted average for Samsung S = 2(9) + 5(7) + 3(4)/2 + 5 + 3 = 6.5 $\frac{2(9)+5(7)+3(4)}{2+5+3}=6.5$

Hence, he should go for Sony J.

DIRECTIONS *for question*: Refer to the graph below and answer the questions that follow.



The graph above gives the IPI (Industrial production Index) for a country. Overall Index is calculated as the weighted sum of the other three areas. The base year is 1981 – 82 (IPI for base year = 100).

Ex. 3: What will be the new overall index for 95-96 (approximately), if the weights of Steel, Coal and Petroleum were to be made equal in 95-96 (i.e. 33.33% for each)?

A. 400 B. 300 C. 330 D. 370

Sol: If for the year 95-96, the weights of Steel, Coal and Petroleum were made equal, then Overall Index would be the simple average of all the three areas, or the weights attached to each would be 1, 1 and 1.

Overall index = 1 × 350 + 1 × 250 + 1 × 400/1 + 1+1

$$\text{Overall index} = \frac{1x350 + 1x250 + 1x400}{1+1+1}$$

Overall index = $\frac{350+250+400}{3} = 333.33$ (or simple average as the weights are equal)

Ex.4: What will be the overall index in a year, if the Steel index, Coal index and Petroleum index are 500, 400 and 200, respectively? Weights in the given year are: 60, 20 and 20, for Steel, Coal and Petroleum, respectively.

A. 290 B. 400 C. 330 D. 420

Sol: Steel index, Coal index and Petroleum index are 500, 400 and 200, respectively.
Weights in the given year are 60, 20 and 20 for Steel, Coal and Petroleum, respectively.
Or, you can say that the Weights in the given year are 3, 1 and 1 for Steel, Coal and Petroleum, respectively.
Thus, Overall index will be calculated as the weighted sum of the other three areas;

$$\frac{3\times 500+1\times 400+1\times 200}{3+1+1} = \frac{2100}{5} = 420$$

With the help of this technique, seemingly tough data-based questions can be solved easily. It is a very useful tool, and can prove to be handy in data as well as in quant questions.

3.1 Data Interpretation: Pie Chart

A PIE CHART is a type of graph in which a circle is divided into sectors, in which each sector represents a proportion of the whole. A *pie chart* looks like – you've guessed it right –a pie! It is a circle with various-sized 'slices' cut out from the center to the edge. You should learn how to make a pie chart, so that you can answer the questions based on the same easily. While making a pie chart every component is graphed in such a manner that the size of the pie it represents, corresponds to its real value on a relative basis i.e. the size of the slices shows the relative size of the categories. Pie charts use values or percentage or angles to show the relative sizes of various categories. In simpler words, the bigger the slice of the pie, the higher is the proportion out of the total that it represents. You must read, understand and apply the following points to solve pie chart problems.

Handling angles, percentages and values

You probably won't be surprised to find that you can solve problems related to pie charts using the properties of a circle and basic concepts of percentages. A whole circle contains 360 degrees. In a pie chart, those 360 degrees corresponds to the total of the values represented in the chart. On the other hand, the distribution might be in terms of percentages, which sums up to 100%.

Now to work with a pie chart, you may need to figure out one of the following three things:

- The size of the angle in a slice.
- The value of a slice.
- The total sum of the values in all the slices.

Pie chart formulas and Important points:

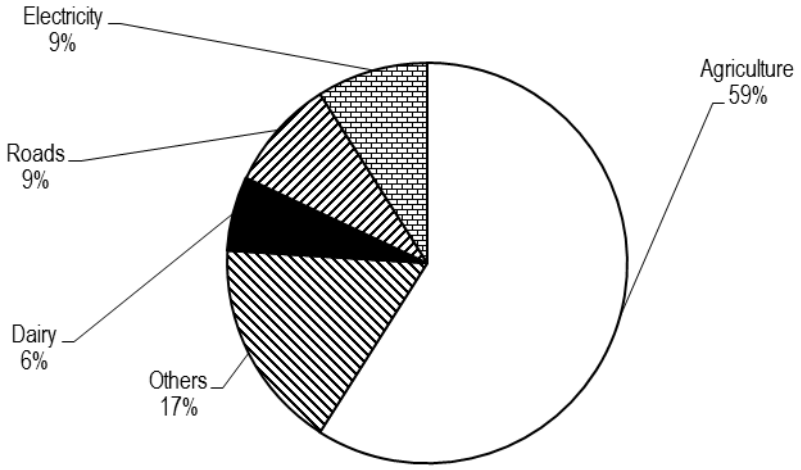
Total angle at the center of a pie chart = 360^0
To convert k% into angle = (k/100) × 360
To convert m^0 into percentage = (m/360) ×100

Before solving the questions of your own, you must understand a few of the pie chart questions and answers. As most of the exams are online these days, so you must practice some pie charts online.

3.2 Pie Charts Examples

DIRECTIONS: Go through the following pie chart and answer the questions given below that.

2000 Haryana State Budget Expenditure (% allocation)



Ex. 1: Approximately how many degrees should there be in the central angle of the sector for agriculture expenditure?

A. 220 B. 213 C. 210 D. 208 E. 198

Sol: In a pie chart, 100% is spread over 360°.
Therefore 1 % = 3.6°. Agriculture expenditure = 59 %. Therefore 3.6 × 59 = 212.4°. Answer is (2)

Ex. 2: Approximately what is the ratio of expenditure on agriculture to that on dairy?

A. 8:3 B. 9:2 C.70:1 D.10:1 E. 6:1

Sol: Over here one common mistake is that students calculate the actual values of agriculture and dairy. Since budget expenditure is proportional to % of area covered, ratio of agriculture to dairy expenditure would be ratio of corresponding % allocations. Therefore, Agriculture/Dairy = 59/6 = 10/1. Answer is (4).

Ex. 3: In Haryana in 2000 a total expenditure of 120mn Rs. was incurred. Approximately how many million did the Haryana government spend on roads?

A. 15.4 B. 20 C. 9 D. 12 E. 10.8

Sol: Total expenditure = 100% = Rs. 120 mn. Expenditure on roads = 9% = 9/100 × 120 = Rs. 10.8mn. Hence the answer is 5th option.

Ex. 4: If Rs. 9mn were spent in 2000 for Dairy what would have been total expense for that year in million?

A. 150 B. 140 C.160 D.130 E. 125

Sol: 9mn were spent for dairy. This represents 6% of total expenditure in the year 2000.

$6 = \frac{\text{Dairy expenditure}}{\text{Total expenditure}} \times 100$; $6 = \frac{9}{\text{Total expenditure}} \times 100$

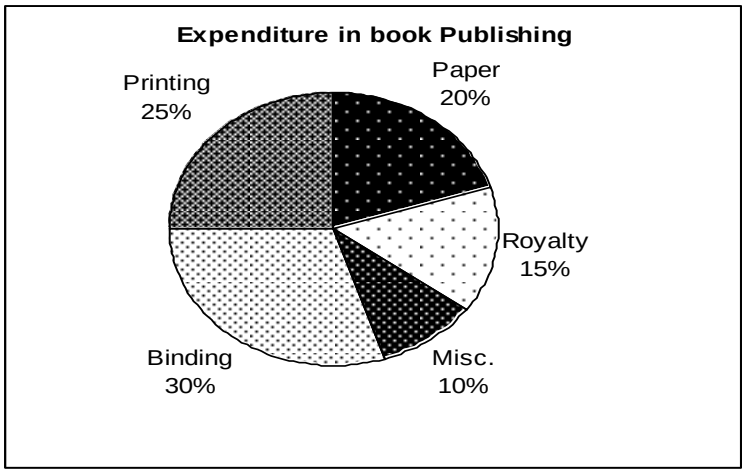
Total Exp = 100 × 9/6 = Rs. 150 mn.

Ex. 5: In 1999, Haryana spent 11% of all its expenses on Roads. "Did Haryana spend more on Roads expenditure in 1999 than in 2000?"to answer the question, we

1. Do not require any additional data.
2. Require to know the total amount spent in each year.
3. Require the exact %age break-up of the various items of expenses in 1999.
4. Require the expenditure on dairy in both the years.

Sol: % gives us just a proportional indication of total quantity.
Unless total expenditure is known, % of that is meaningless.
Hence to compare expenditure on roads in 1999 and 2000. Total amount spent must be known.
Answer is (2).

DIRECTIONS *for questions 6 to 10*: *The following pie-diagram shows the expenditure incurred in the preparation of a book by a publisher, under various heads.*



Ex. 6: What is the angle of pie- diagram showing the expenditure incurred on paying the royalty?

A. 24 deg B. 48 deg C. 54 deg D. 15 deg

Sol: 15% of 360^0 = $\frac{15 \times 360}{100} = 54^\circ$

Ex. 7: The marked price of a book is 20% more than the C.P. If the marked price of the book be Rs. 30, what is the cost of paper used in a single copy of the book?

A. Rs.6 B. Rs.5 C. Rs. 4.50 D. Rs. 6.50

Sol: The marked price = Rs.30/-. The cost price = $\frac{30}{1.2} = 25$

Cost of paper = 20% of 25 = 0.2×25= Rs 5.

Ex. 8: Which two expenditures together will form an angle of 108 degrees at the centre of the pie-diagram?

A. Paper & Royalty B. Printing & Royalty C. Paper & Misc. D. None of these

Sol: Going by the options, paper and misc. constitute (20%+10%) = 30% of 360^0 = 108^0.

Ex. 9: If the difference between two expenditures be represented by 18 degrees in the diagram (if the diagram is made in terms of degrees) these expenditures are

A. Printing & Royalty B. Paper & Binding C. Printing & Misc. D. None of these

Sol: For printing and royalty = (25%-15%)=10%=36^0
For paper and binding = (30%-20%)=10%=36^0
For printing and misc = (25%-10%)=15%=54^0.
Hence none of the options satisfy.

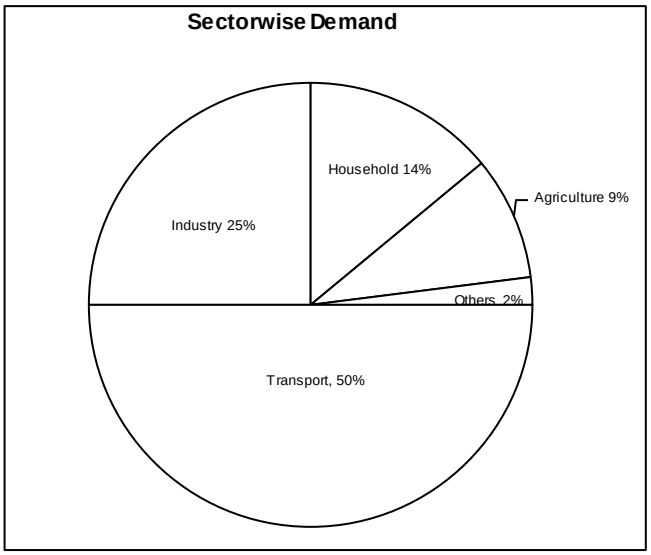
Ex. 10: Royalty is less than paper charges by

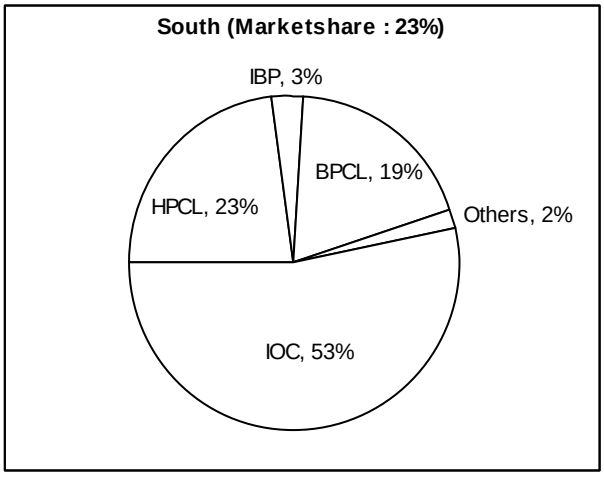
A. 25% B. 27 2/3% C. 23.5% D. 20%

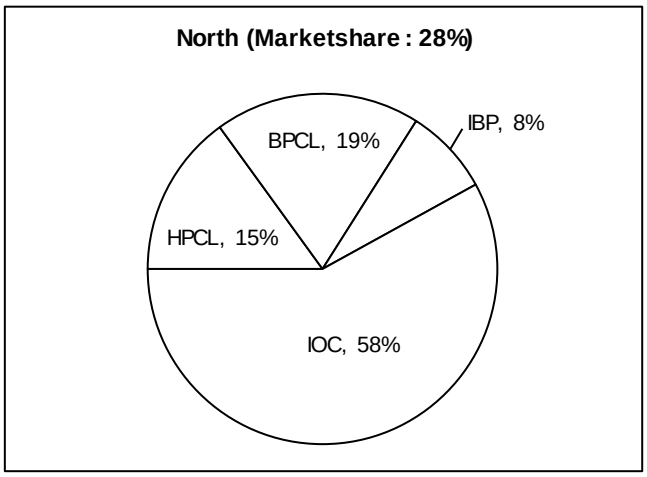
Sol: Royalty is 15% of the total cost and paper is 20% of the total cost.
So the required percentage = $\frac{20-15}{20}\times 100=\frac{5}{20}\times 100=25\%$.

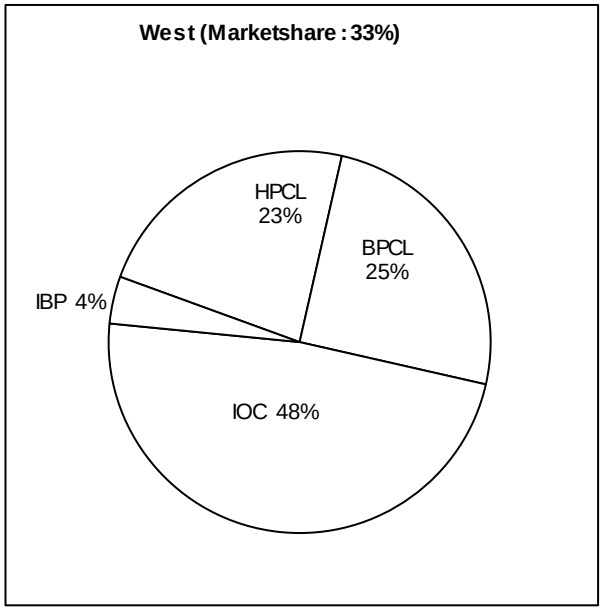
DIRECTIONS *for questions 11 - 15:* Refer to the pie charts given below and answer the questions that follow.

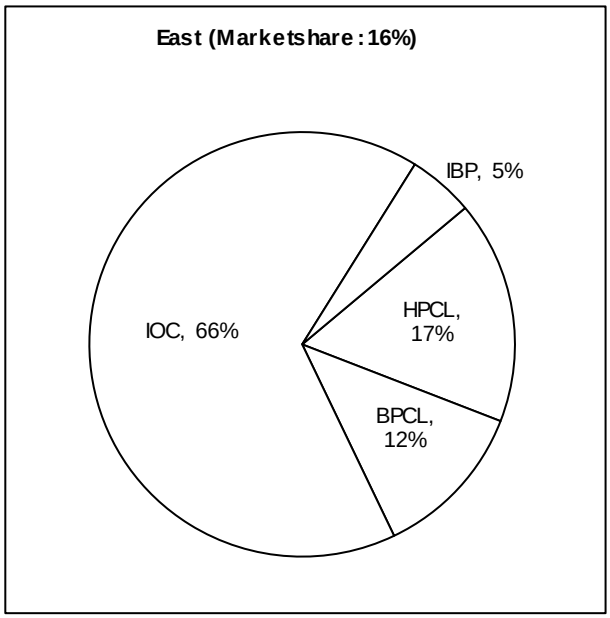
YEAR 1995
REGIONAL MARKET SHARES











The pie charts above give the sales wise distribution (Market shares) of the major oil producing giants in India. The Total sales for these major oil giants (all over India) in 1995 were $ 600 bn (where $ 1bn = Rs. 3,300 crores). The total demand for oil in 1995 was 100,000 crore kgs (all India).

Ex. 11: It can be inferred from the graphs that in 1995,

A. IOC is the largest oil company in India.
B. HPCL is the smallest oil company in India.
C. The greatest consumption of oil is by the 'agriculture' sector.
D. None of the above.

Sol: From the pie charts, we can see that IOC had more than 50% share in all regions. So, we can safely say that IOC is the largest company.

Ex. 12: If total sales (in kg), all over India, by the given oil giants in 1995 is 80% of the total demand, the price of oil (on an average) in 1995 must have been

A. Rs. 18.2 / kg B. Rs. 13.6 / kg C. Rs. 15.84 / kg D. None of these

Sol: Total sales = $ 600 bn. Total demand = 100000 crore kgs.
Total sales = 80% of 100000 = 80000 crorekgs.
Price of oil = Rs. (600 × 3300 crore) / 80000 crorekgs
= Rs. 24.75 /kg. So answer is 4th option.

Ex. 13: The total all India sales of IOC were approximately

A. $ 300 bn B. $ 310 bn C.$ 331.2 bn D. $ 340.1 bn

Sol: Directly calculate from all the four graphs and add.

Total Sales of IOC= $ 600 (0.53×0.23+0.28×0.58+0.48 ×0.33+ 0.16 ×0.66) bn
= 600 (0.1219+0.1624+0.1584+0.1056)
=$328.9 bn = bn$329 (Approx.)
The answer is 3rd option.

Ex. 14: The sales of HPCL in West exceeded the sales of BPCL in North by

A. Rs. 3,397crores B. Rs. 44946 crores C. Rs. 5,479 crores D. $ 4.3 bn

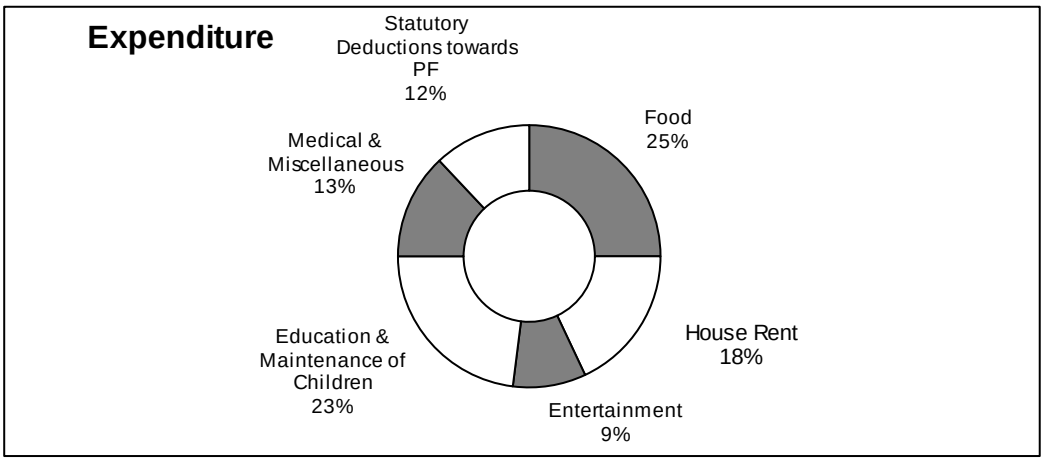
Sol: Sales of HPCL in West = $ 600bn × 0.33 × 0.23 = $ 45.54 bn.
Sales of BPCL in North = $ 600bn × 0.28 × 0.19 = $ 31.92 bn.
Difference = $ 13.62bn = Rs. 13.254 × 3300 crore = Rs. 44946 crore.
So answer is 2nd option.

Ex. 15: The total all India sales of all the given companies other than IOC in 1996 was

A. $ 268.8 bn B. $ 372.9 bn C. $ 280 bn D. Cannot be determined

Sol: For 1996 we don't have any data. So answer is 4th option.

DIRECTIONS *for questions 16 – 19:* *Analyze the following pie chart and answer the questions given afterwards. Given is the domestic budget of family X. The total earnings of family X are Rs. 3,365 per month basic, plus 10% as house rent subsidy on the monthly salary.*



Ex. 16: If family X pays 18% of its total emoluments as house rent, calculate the amount of house rent per month it is paying.

A. Rs. 618 B. Rs. 600 C. Rs. 666 D. Rs. 675

Sol: Total Emoluments = 3365 + 10% of 3365 = Rs. 3701.5
18% of 3701.5 = Rs. 666

Ex. 17: The statutory deduction towards provident fund @ 12% per month of the total salary works out to be Rs. _____ per annum.

A. 4,854 B. 5,330.16 C. 53.85 D. 4,485.6

Sol: Provident Fund deduction of Total Salary = 12% of Rs. 3701.5 = Rs. 444.18
Deductions per annum = Rs. 444.18 × 12 = Rs. 5330.16.

Ex. 18: After provident fund deduction and payment of house rent, what is the approximate take home pay packet of family X (in Rupees)?(Use data from the Question 6)

A. 2,631 p.m. B. 2,591 p.m. C. 2,355 p.m. D. 2,350 p.m.

Sol: Provident Fund Deduction = Rs. 444.18
House Rent = Rs. 666.27. Total = Rs. 1110.45
Take Home Pay Packet = Rs. 3701.5 – 1110.45 = Rs. 2591.05

Ex. 19: What is the total amount per month the family spends on food and entertainment expenses?

A. Rs. 1,100 B. Rs. 1,259 C. Rs. 1,104 D. Rs. 1,140

Sol: Expenditure on Food = Rs. 925.37
Expenditure on Entertainment = Rs. 333.14 $\Rightarrow$ Total on both = 1258.51

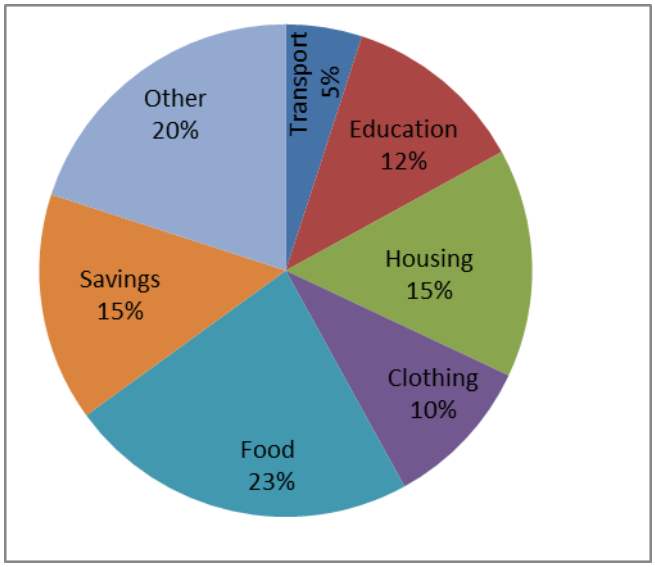
Important points:

Total angle at the center of a pie chart = 360^0
To convert k% into angle = (k/100) × 360
To convert m^0 into percentage = (m/360) ×100

3.3 Illustration

DIRECTIONS: *The circular graph given below shows the spending by a family on various items during the year 1998. Study the graph and answer the questions that follow.*

(Percent of money spent by a family during 1998)



Ex. 1: If the total amount spent during the year 1998 was Rs. 46000, how much was the amount spent on food?

Sol: Food comprises of 23% of the total spending, therefore, amount spent on food = 46000 × 23/100 = Rs 10580.

Ex. 2: If the total amount spent during the year 1998 was Rs. 46000, how much was the amount spent on clothing and housing together?

Sol: Amount spent on clothing and housing are 10% and 15%, respectively. Therefore, total amount spent is, 46000 × 10/100 + 46000 × 15/100 = = (4600 + 6900) = 11500

Note: Instead of finding the two values individually, you can sum up the percentages spent on clothing and housing i.e. (10 + 15) i.e. 25 %. Now, the total amount spent on clothing and housing together is 46000 × 25/100 = 11500

Ex. 3: What was the ratio of the total amount of money spent on housing to that spent on education?

Sol: In this question too, there is no need to find out the amounts spent on housing or education. Instead you can get the answer by comparing the percentages i.e. 15 : 12 ⇒ 5 : 4.

Ex. 4: As per the chart, the maximum amount was spent on which item, if the total expenditure was given to be Rs. 48256?

Sol: Irrespective of the total expenditure, the item on which maximum amount was spent will be the item on which maximum percentage of total was spent, i.e. food (23% spent).

Ex. 5: The amount spent on Housing is how many times of the amount spend on transport, if the total expenditure happens to be Rs. 83000?

Sol: You can avoid doing long calculations in this question as well. In terms of percentage, housing and transport are 15% and 5% respectively. Hence the expenditure on housing will be 15/5 = 3 times of the percentage on transport.